

Introduction to Data Science and Engineering

- R programming

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University of Hong Kong

Some slides retrofitted by Michael Wu



Some materials courtesy of
Rafael A. Irizarry, and are modified
from the original version.



R markdown for stylish texts, inline code and results

Assignments are handed-out and received as R markdown format, so please get familiar with it

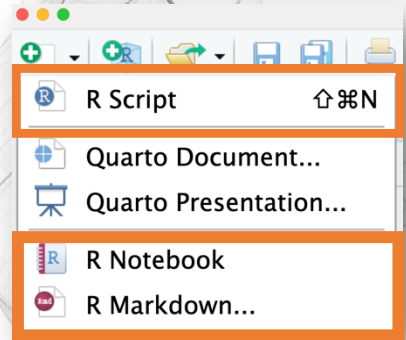
Markdown language

- **Markdown is a lightweight markup language for creating formatted text using a plain-text editor.** John Gruber and Aaron Swartz created Markdown in 2004 as a markup language that is appealing to human readers in its source code form. **Markdown is widely used in blogging, instant messaging, online forums, collaborative software, documentation pages, and readme files**

<https://en.wikipedia.org/wiki/Markdown>

Markdown	
	
Internet media type	<code>text/markdown</code> ^[1]
Uniform Type Identifier (UTI)	<code>net.daringfireball.markdown</code>
Developed by	John Gruber and Aaron Swartz
Initial release	March 19, 2004 (17 years ago) ^{[2][3]}
Latest release	1.0.1 (December 17, 2004 (16 years ago) ^[4])
Type of format	Markup language
Extended to	pandoc, MultiMarkdown, Markdown Extra, CommonMark, ^[5] RMarkdown ^[6]
Open format?	Yes ^[7]
Website	daringfireball.net/projects/markdown/ 

R Markdown / R Notebook



- **R Markdown (.Rmd):** for creating polished, reproducible documents like reports or tutorials.
 - Codes are executed when *knitted*
- **R Notebook:** an interactive extension of R Markdown, ideal for exploratory analysis, prototyping, and development.
 - Codes are executed in real-time with inline results.
- **R Scripts (.R):** plain text files used to write and execute R code, no additional formatting or documentation features.
 - Codes are run in an R console

A TINY CHEATSHEET ON



MARKDOWN

Markdown is a lightweight markup language. Its design allows it to be converted to many output formats like HTML

Titles

Heading 1

Heading 2

Heading 3

Heading 4

Heading 5

Heading 6

Text

Italic

Also Italic

****Bold****

Also Bold

Lists

* Item 1

* Item 2

- This works

- Too

1. Ordered

2. Lists

3. Are Fun

Links & Images

![Image Title](Image URL)

[Click Me](Link URL)

[profile](github.com/godcrampy)

Quotes

Linus Said:

```
> Cheat Sheets are Cool!  
And they are Funky
```

Checklist

- [x] this is a complete item

- [] this is an incomplete item

Code

```
```javascript  
function foo() {
 // Comment
 let a = 5;
 let b = 7;
 return a + b;
}
```\n`<inline-code>`
```

Markdown Cheatsheet

Work with Markdown

- **Plenty of online resources:**

- <https://stackedit.io/>
- <https://markdownlivepreview.com/>

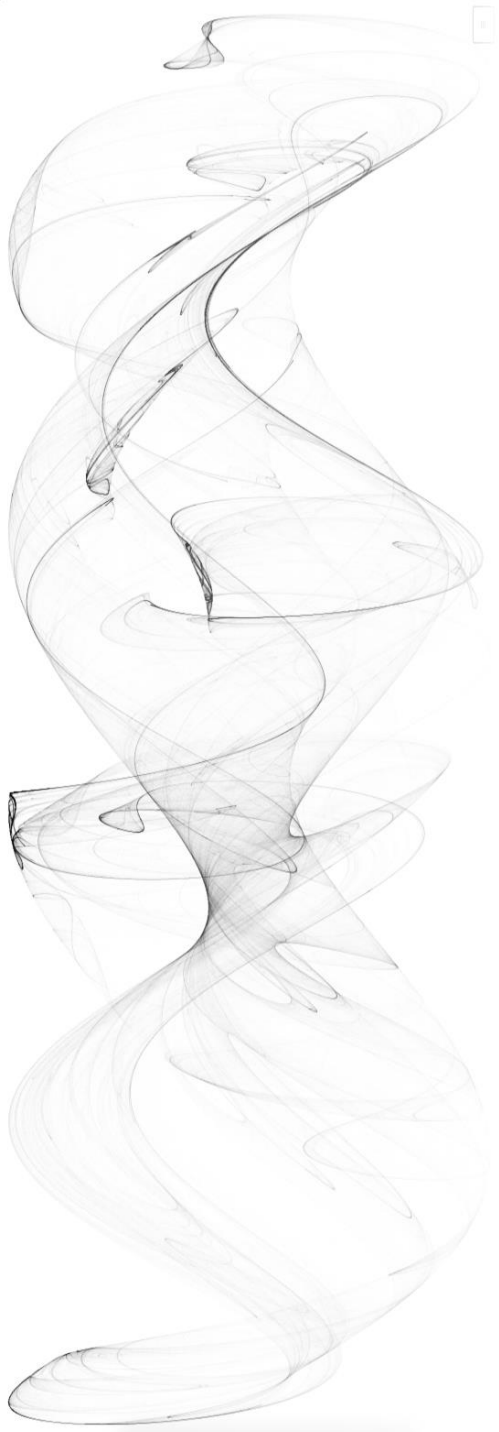
ChatGPT 5 ▾

↑ Share ...

List me some resources for learning and writing markdwns online, put them in a table please.

Here's a table of popular resources you can use to learn and practice writing Markdown online:

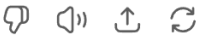
Resource	Type	Key Features	URL
Markdown Guide	Tutorial & Reference	Comprehensive syntax guide, extended syntax, cheat sheet	https://www.markdownguide.org ↗
GitHub Markdown Guide	Tutorial & Reference	GitHub Flavored Markdown (GFM) documentation, examples	https://docs.github.com/en/get-started/writing-on-github ↗
Dillinger	Online Editor	Live preview, export to HTML/PDF, cloud integration	https://dillinger.io ↗
StackEdit	Online Editor	WYSIWYG + Markdown hybrid, offline mode, Google Drive sync	https://stackedit.io ↗



Here’s a table of popular resources you can use to learn and practice writing Markdown online:

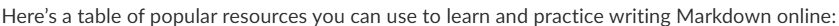
Resource	Type	Key Features	URL
Markdown Guide	Tutorial & Reference	Comprehensive syntax guide, extended syntax, cheat sheet	https://www.markdownguide.org ↗
GitHub Markdown Guide	Tutorial & Reference	GitHub Flavored Markdown (GFM) documentation, examples	https://docs.github.com/en/get-started/writing-on-github ↗
Dillinger	Online Editor	Live preview, export to HTML/PDF, cloud integration	https://dillinger.io ↗
StackEdit	Online Editor	WYSIWYG + Markdown hybrid, offline mode, Google Drive sync	https://stackedit.io ↗
HackMD	Collaboration Tool	Real-time collaborative Markdown editing, diagrams, integrations	https://hackmd.io ↗
Obsidian Help	App Documentation	Guides for using Markdown in Obsidian note-taking	https://help.obsidian.md ↗
Markable	Online Editor	Instant HTML preview, export options, simple UI	https://markable.in ↗
Learn Markdown in Y Minutes	Quick Reference	Concise syntax overview with examples	https://learnxinyminutes.com/docs/markdown ↗

Do you want me to also **add a quick “Markdown syntax cheat sheet”** to go with this table so you can reference common formatting instantly?

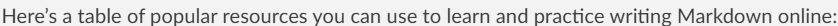


+ Ask anything





Do you want me to also ****add a quick “Markdown syntax cheat sheet”**** to go with this table so you can reference common formatting instantly?



Do you want me to also **add a quick “Markdown syntax cheat sheet”** to go with this table so you can reference common formatting instantly?

Markdown language in Jupyter Notebook

The image displays a Jupyter Notebook interface with a Markdown cell. The cell content is as follows:

```
#ColabFold: AlphaFold2 w/ MMseqs2
```

- **21Aug2021: The MSA/Templates issues should now be resolved! Please report any errors you see.**

 ``

Easy to use AlphaFold2 [(Jumper et al. 2021)](<https://www.nature.com/articles/s41586-021-03819-2>) protein structure prediction using multiple sequence alignments generated through an MMseqs2 API. For details, refer to our manuscript:

[Mirdita M, Ovchinnikov S, Steinegger M. ColabFold - Making protein folding accessible to all. *bioRxiv*, 2021](<https://www.biorxiv.org/content/10.1101/2021.08.15.456425v1>)

- This notebook provides basic functionality, for more advanced options (such as modeling heterocomplexes, increasing recycles, sampling, etc.) see our [advanced notebook](https://colab.research.google.com/github/sokrypton/ColabFold/blob/main/beta/AlphaFold2_advanced.ipynb).
- This notebook replaces the homology detection of AlphaFold2 with MMseqs2. For a comparison against the [Deepmind Colab](<https://colab.research.google.com/github/deepmind/alphafold/blob/main/notebooks/AlphaFold.ipynb>) and the full [AlphaFold2](<https://github.com/deepmind/alphafold>) system read our [preprint](<https://www.biorxiv.org/content/10.1101/2021.08.15.456425v1>).

For more details, see [bottom](#Instructions) of the notebook and checkout the [ColabFold GitHub](<https://github.com/sokrypton/ColabFold>).

The right sidebar shows the rendered HTML output of the Markdown cell:

ColabFold: AlphaFold2 w/ MMseqs2

- **21Aug2021: The MSA/Templates issues should now be resolved! Please report any errors you see.**

Easy to use AlphaFold2 (Jumper et al. 2021) protein structure prediction using multiple sequence alignments generated through an MMseqs2 API. For details, refer to our manuscript:

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A cartoon cat character is visible in the bottom right corner of the rendered output.

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```

- **21Aug2021: The MSA/Templates issues should now be resolved! Please report any errors you see.**

```

```

Easy to use AlphaFold2 [(Jumper et al. 2021)](<https://www.nature.com/articles/s41586-021-03819-2>) protein structure prediction using multiple sequence alignments generated through an MMseqs2 API. For details, refer to our manuscript:

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- 

For more details, see [bottom](#) of the notebook and checkout the [ColabFold GitHub](#).



Download the example

- Download the 01.ColabFold.Rmd file from Moodle to your Desktop or elsewhere
- Open 01.ColabFold.Rmd in R Studio, knit it

ColabFold: AlphaFold2 using MMseqs2

Author: Google Brain

Year: 2021

OK, google again

Easy to use AlphaFold2 ([Jumper et al. 2021](#)) protein structure prediction using multiple sequence alignments generated through an MMseqs2 API. For details, refer to our manuscript:

Mirdita M, Ovchinnikov S, Steinegger M. ColabFold - Making protein folding accessible to all. *bioRxiv*, 2021

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Titles

```
▼ #ColabFold: AlphaFold2 using MMseqs2
▼ ##Author: Google Brain
▼ ###Year: 2021
▼ ####OK, google again
```

ColabFold: AlphaFold2 w/ MMseqs2

Author: Google Brain

Year: 2021

OK, google again

A TINY CHEATSHEET ON



MARKDOWN

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Titles

```
# Heading 1
## Heading 2
### Heading 3
#### Heading 4
##### Heading 5
##### Heading 6
```

Text

```
*Italic*
_Also Italic_
**Bold**
_Also Bold_
```

Lists

```
* Item 1
* Item 2

- This works
- Too

1. Ordered
2. Lists
3. Are Fun
```

Links & Images

```
(Image URL)
[Click Me](Link URL)
[profile](github.com/godcrampy)
```

Quotes

```
Linus Said:

> Cheat Sheets are Cool!
And they are Funky
```

Checklist

```
- [x] this is a complete item
- [ ] this is an incomplete item
```

Code

```
```javascript
function foo() {
 // Comment
 let a = 5;
 let b = 7;
 return a + b;
}
...
<inline-code>
```

# Split line



```
▼ #ColabFold: AlphaFold2 using MMseqs2
▼ ##Author: Google Brain
▼ ###Year: 2021
▼ ####OK, google again
```

```
▼ -----
```



Actually, three hyphens  
are enough “---”

ColabFold: AlphaFold2 w/ MMseqs2

Author: Google Brain

Year: 2021

OK, google again





# Line break is counter-intuitive in markdown

```
ColabFold: AlphaFold2 using MMseqs2
Author: Google Brain
Year: 2021
Hello
World

OK, google again
```

**ColabFold: AlphaFold2 using MMseqs2**

**Author: Google Brain**

**Year: 2021**

Hello World

OK, google again



# Add two spaces to the end of a line as a newline

ColabFold: AlphaFold2 w/ MMseqs2

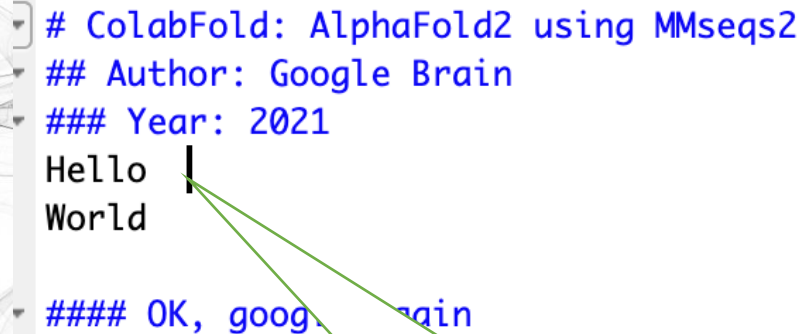
Author: Google Brain

Year: 2021

Hello

World

OK, google again



```
ColabFold: AlphaFold2 using MMseqs2
Author: Google Brain
Year: 2021
Hello
World
OK, google again
```

Two spaces here work



# But "two spaces" does not work in titles

```
ColabFold: AlphaFold2 using MMseqs2
Author: Google Brain
Year: 2021 |
OK, google again
```

Two spaces here  
don't work

ColabFold: AlphaFold2 w/ MMseqs2

Author: Google Brain

Year: 2021

OK, google again

# Use a special html tag to break the line

```
ColabFold: AlphaFold2 using MMseqs2
Author: Google Brain
Year: 2021

OK, google again
```

ColabFold: AlphaFold2 w/ MMseqs2

Author: Google Brain

Year: 2021

OK, google again

<br> means a newline in html,  
And don't forget a newline after <br>

# Italic font

Easy to use *AlphaFold2* [(Jumper et al. 2021)](<https://www.nature.com/articles/s41586-021-03819-2>) protein structure prediction using multiple sequence alignments generated through an `_MMseqs2_` API. For details, refer to our manuscript:

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A TINY CHEATSHEET ON



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## Titles

# Heading 1  
## Heading 2  
### Heading 3  
#### Heading 4  
##### Heading 5  
##### Heading 6

## Text

*\*Italic\**  
*\_Also Italic\_*  
**\*\*Bold\*\***  
**\_Also Bold\_**

## Lists

- \* Item 1
- \* Item 2

- This works
- Too

1. Ordered
2. Lists
3. Are Fun

## Links & Images

!([Image Title])(Image URL)  
[Click Me](Link URL)  
[profile](github.com/godcrampy)

## Quotes

Linus Said:

> Cheat Sheets are Cool!  
And they are Funky

## Checklist

- [x] this is a complete item
- [ ] this is an incomplete item

## Code

```
```javascript
function foo() {
  // Comment
  let a = 5;
  let b = 7;
  return a + b;
}
...
<inline-code>
```

Bold font

Easy to use **AlphaFold2** [(Jumper et al. 2021)](<https://www.nature.com/articles/s41586-021-03819-2>) protein structure prediction using multiple sequence alignments generated through an **MMseqs2** API. For details, refer to our manuscript:

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Also Italic

*****Bold*****

Also Bold

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```
```javascript
function foo() {
 // Comment
 let a = 5;
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 return a + b;
}
...
<inline-code>
```

# List items with bullet

```
- This notebook provides basic functionality, for more advanced options (such as modeling heterocomplexes, increasing recycles, sampling, etc.) see our [advanced notebook](https://colab.research.google.com/github/sokrypton/ColabFold/blob/main/beta/AlphaFold2_advanced.ipynb).
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```

One space here

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```javascript  
function foo() {  
  // Comment  
  let a = 5;  
  let b = 7;  
  return a + b;  
}  
...  
`<inline-code>`
```

List items with number

1. This notebook provides basic functionality, for more advanced options (such as modeling heterocomplexes, increasing recycles, sampling, etc.) see our [advanced notebook](https://colab.research.google.com/github/sokrypton/ColabFold/blob/main/beta/AlphaFold2_advanced.ipynb).
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```javascript
function foo() {
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 let b = 7;
 return a + b;
}
...
<inline-code>
```

# Link

Syntax: [text to show](the link)

Easy to use *\*AlphaFold2\** [(Jumper et al. 2021)](<https://www.nature.com/articles/s41586-021-03819-2>) protein structure prediction using multiple sequence

*AlphaFold2* ([Jumper et al. 2021](#)) proteir



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```
```javascript
function foo() {
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  let a = 5;
  let b = 7;
  return a + b;
}
...
<inline-code>
```

Image

Syntax: ![empty or a figure description](the image link)

``



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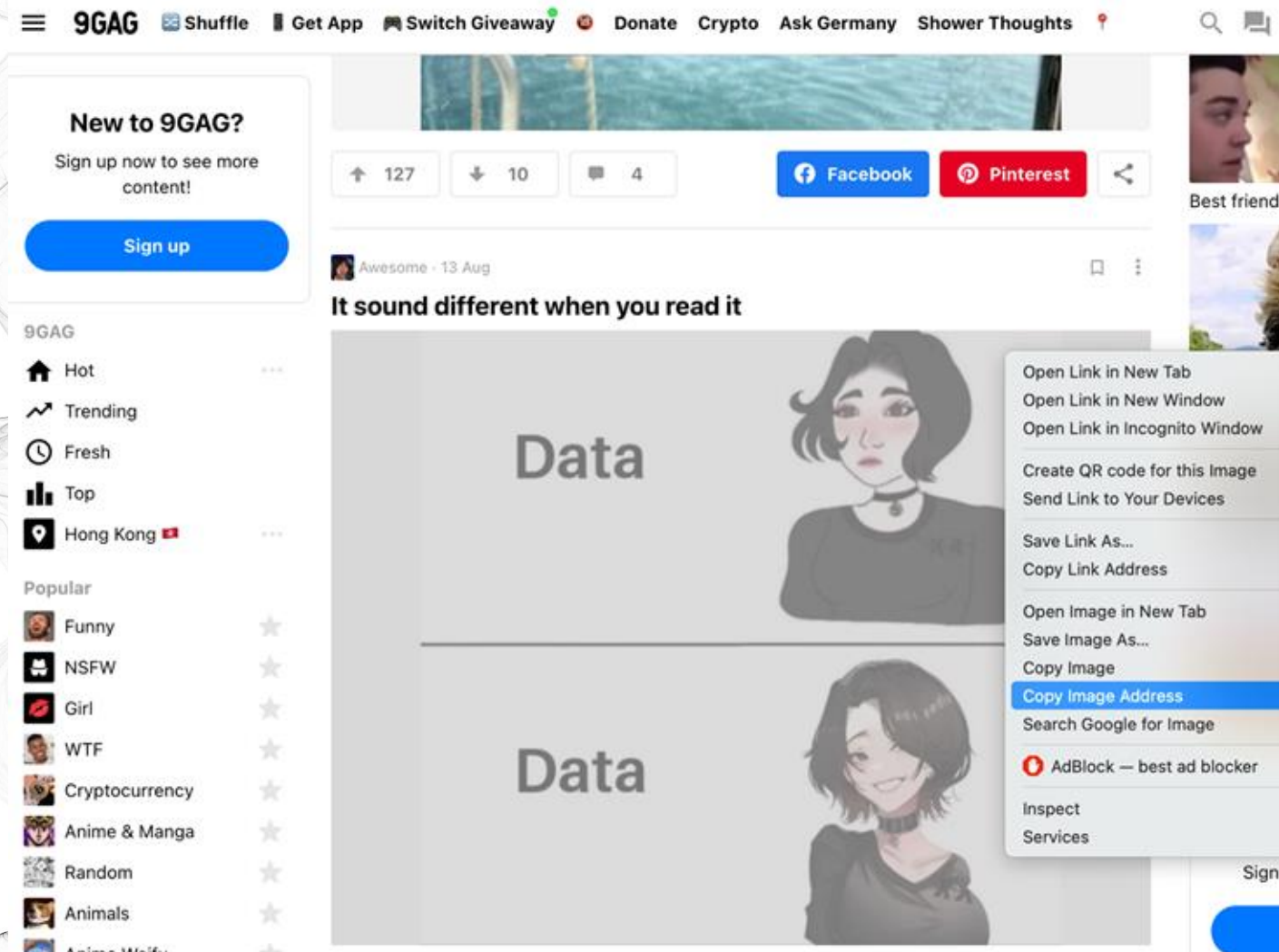
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```
```javascript
function foo() {
 // Comment
 let a = 5;
 let b = 7;
 return a + b;
}
...
<inline-code>
```

# Copy the link of an image online



)  
Easy to use *\*AlphaFold2\** [(Jumper et al. 2021)](<https://www.nature>

- 21Aug2021: The MSA/Templates issues should now be resolved!

Data



Data



Easy to use AlphaFold2 (Jumper et al. 2021) protein structure prediction



**Bob E. Hayes**

@bobehayes



How do you say the word, "data?" [#datascience](#) [#bigdata](#)

**day-tuh**

**63%**

dat-uh

20.5%

dah-tuh

16.4%

146 votes · Final results

3:13 AM · Nov 25, 2015



17



3



Copy link to Tweet

[Tweet your reply](#)

# Quote

```
> Easy to use *AlphaFold2* [(Jumper et al. 2021)](https://www.nature.com/articles/s41586-021-03819-2) protein structure prediction using multiple sequence alignments generated through an _MMseqs2_ API. For details, refer to our manuscript:
```

Easy to use *AlphaFold2* ([Jumper et al. 2021](#)) protein structure prediction using multiple sequence alignments generated through an *MMseqs2* API. For details, refer to our manuscript:

Depends on the platform you are using, it could be a grey background, or a grey bar at the front



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  let b = 7;  
  return a + b;  
}  
...  
<inline-code>
```

Checklist

"- [] ": -, space, [, space,], space

"- [x] ": -, space, [, x,], space (it has to be x)

```
- [x] This notebook provides basic functionality, for more advanced options
(such as modeling heterocomplexes, increasing recycles, sampling, etc.) see our
[advanced notebook](https://colab.research.google.com/github/sokrypton/
ColabFold/blob/main/beta/AlphaFold2_advanced.ipynb).
- [ ] This notebook replaces the homology detection of AlphaFold2 with MMseqs2.
For a comparison against the [Deepmind Colab](https://colab.research.google.
com/github/deepmind/alphafold/blob/main/notebooks/AlphaFold.ipynb) and the full
[AlphaFold2](https://github.com/deepmind/alphafold) system read our [preprint]
(https://www.biorxiv.org/content/10.1101/2021.08.15.456425v1).
```

- ☒ This notebook provides basic functionality, for more advanced options (such as modeling heterocomplexes, increasing recycles, sampling, etc.) see our [advanced notebook](#).
- ☐ This notebook replaces the homology detection of AlphaFold2 with MMseqs2. For a comparison against the [Deepmind Colab](#) and the full [AlphaFold2](#) system read our [preprint](#).



Markdown is a lightweight markup language. Its design allows it to be converted to many output formats like HTML

Titles

Heading 1
Heading 2
Heading 3
Heading 4
Heading 5
Heading 6

Text

Italic
Also Italic
Bold
Also Bold

Lists

- * Item 1
- * Item 2

- This works
- Too

1. Ordered
2. Lists
3. Are Fun

Links & Images

![Image Title](Image URL)
[Click Me](Link URL)
[profile](github.com/godcrampy)

Quotes

Linus Said:

> Cheat Sheets are Cool!
And they are Funky

Checklist

- [x] this is a complete item
- [] this is an incomplete item

Code

```
```javascript
function foo() {
 // Comment
 let a = 5;
 let b = 7;
 return a + b;
}
...
<inline-code>
```

# Inline code

```: backtick, the key left to 1



```
> Easy to use `AlphaFold2` [(Jumper et al. 2021)](https://www.nature.com/articles/s41586-021-03819-2) protein structure prediction using multiple sequence alignments generated through an `MMseqs2` API. For details, refer to our manuscript:
```

Easy to use AlphaFold2 ([Jumper et al. 2021](#)) protein structure prediction using multiple sequence alignments generated through an MMseqs2 API. For details, refer to our manuscript:

A TINY CHEATSHEET ON



MARKDOWN

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# Code block

`: backtick, the key left to 1



```
print("Hello world")
```

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```



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### Heading 3  
#### Heading 4  
##### Heading 5  
##### Heading 6

## Text

*\*Italic\**  
*Also Italic*  
**\*\*Bold\*\***  
**Also Bold**

## Lists

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}
```
```

<inline-code>

# R markdown code block specialties

```
23
24 ▾ ```{r block1}
25 summary(cars)
26 ▴ ```
27
```

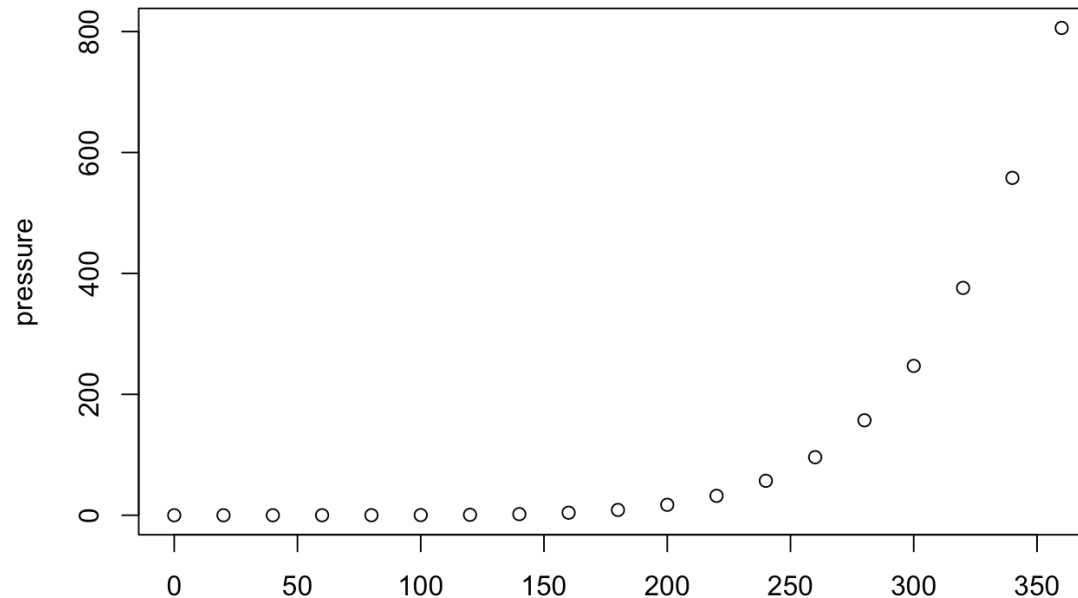
Run the code block,  
show code and results

```
summary(cars)
```

```
speed dist
Min. : 4.0 Min. : 2.00
1st Qu.:12.0 1st Qu.: 26.00
Median :15.0 Median : 36.00
Mean :15.4 Mean : 42.98
3rd Qu.:19.0 3rd Qu.: 56.00
Max. :25.0 Max. :120.00
```

```
28 ▾ ```{r block2, echo=FALSE}
29 plot(pressure)
30 ▴ ```
31
```

Run the code block,  
show only the results



# Tables

| Left-aligned | Center-aligned | Right-aligned |
|--------------|----------------|---------------|
| git status   | git status     | git status    |
| git diff     | git diff       | git diff      |

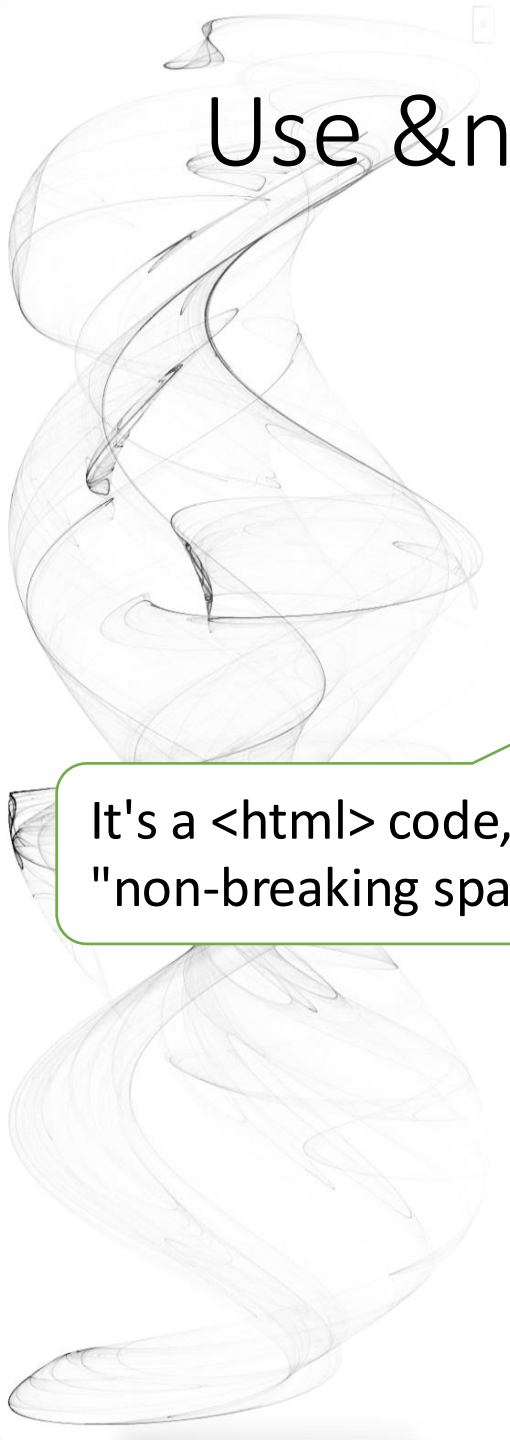
| Left-aligned | Center-aligned | Right-aligned |
|--------------|----------------|---------------|
| git status   | git status     | git status    |
| git diff     | git diff       | git diff      |



| Left-aligned | Center-aligned | Right-aligned |
|--------------|----------------|---------------|
| git status   | git status     | git status    |
| git diff     | git diff       | git diff      |

What if I really want to put a space before git?

You can create a table using this online markdown table code generator: [https://www.tablesgenerator.com/markdown\\_tables](https://www.tablesgenerator.com/markdown_tables)



Use &n

It's a `<html>` code,  
"non-breaking spa

```
Left-aligned	Center-aligned	Right-aligned

git status git status git status
 git diff git diff git diff
```

It's a `<html>` code, means "non-breaking space"

| Left-aligned | Center-aligned | Right-aligned |
|--------------|----------------|---------------|
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